



Ali R. Vahdati

EVOLUTIONARY BIOLOGIST · COMPUTATIONAL BIOLOGIST

Zurich, Switzerland

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Key Skills

Scientific Modeling

Proficient in creating and interpreting complex biological models to understand and predict biological phenomena, contributing to advancements in fields such as human evolution, population genetics, and ecology.

Machine Learning

Applied machine learning and deep learning techniques to understand large datasets and make predictions.

Statistics

Utilized advanced statistical methods to interpret and analyze data from various biological research projects, facilitating informed decision-making.

Mathematical Optimization

Employed mathematical optimization techniques, such as linear programming, in biological modeling and data analysis, optimizing solutions for complex biological processes.

Agent-based Modeling

Developed agent-based models to simulate and analyze complex biological systems, contributing to a deeper understanding of phenomena such as human genetic diversity and ancient human dispersals.

Managing Large Datasets

Experienced in handling and analyzing large genomic datasets, facilitating the extraction of meaningful insights and advancing research in evolutionary biology.

Population Genetics

Applied knowledge of population genetics in various research projects, contributing to the understanding of genetic diversity and evolution.

Programming

Proficient in multiple programming languages and tools including Julia, Python, R, Nim, LaTeX, MySQL, SQLite, Git, Continuous integration, Unit testing, and Docker, enabling the development of efficient algorithms and models for biological research.

Data Visualization

Skilled in using tools like Vega-lite, Matplotlib, and Makie.jl, as well as visualization principles to create effective visual representations of complex data, aiding in data interpretation and communication of research findings.

Soft Skills

Demonstrated resilience, leadership, commitment, and adaptability in various research and teaching roles, contributing to successful project outcomes and effective team collaborations.

Transferable Skills

Proven ability in teaching, presentation, writing, time management, and project management, enhancing the effectiveness of research processes and facilitating knowledge transfer in academic settings.

Languages

English (fluent), Persian (native), German (B1)

Work Experience

Consultant

IBS center for climate physics, University of Pusan

Busan, South Korea

May and June 2022

- Developed a comprehensive model to analyze the impact of climate on human genetic diversity. This model is being used for multiple projects.

Senior lecturer

Christoph Zollikofer group, Department of Anthropology, University of Zurich

Zurich, Switzerland

Aug. 2020 - Current

- Conducted extensive research on ancient human dispersals and extinctions using advanced simulation techniques. My work has provided valuable insights into human evolutionary history.

Postdoctoral researcher

Zurich, Switzerland

Christoph Zollikofer group, Department of Anthropology, University of Zurich

Sep. 2017 - Aug. 2020

- Led multiple research projects, utilizing statistical tools and machine learning techniques to analyze and interpret data. My work resulted in several significant findings, contributing to the broader field of anthropology and computational biology.

Modeler

Madeira, Portugal

University of Madeira

2022

- Constructed a model for understanding metacommunity dynamics of complex life-cycles in exploited seascapes.

Modeler

Mérida, Mexico

Centro de investigación Científica de Yucatán

Oct. 2021

- Developed statistical models to understand interaction strengths in food webs.

Education

University of Zurich

Zurich, Switzerland

PhD in Evolutionary Biology

Jan. 2012 - Sep. 2017

- Advisor: Prof. Dr. Andreas Wagner, Institute of Evolutionary Biology and Environmental Studies.
- Thesis: Genotype networks: convergent evolution, population size and adaptation.
- Published four scholarly papers.

University of Manchester

Manchester, UK

MSc in Evolutionary Biology and Genomics

Sep. 2010 - Sep. 2011

- Advisor 1: Prof. Dr. Reinmar Hager, Faculty of Life Sciences.
- Thesis 1: Identifying QTL and genomic imprinting effects on mice tissue phenotypes.
- Advisor 2: Prof. Dr. Casey Bergman, Faculty of Life Sciences.
- Thesis 2: Analysis of replication timing on genetic variation in the human genome.
- Published one scholarly paper.

Selected Peer-reviewed Papers

EvoDynamics.jl: a framework for modeling eco-evolutionary dynamics

Journal of Open Source Software

R Vahdati A, Melián C J

2022

Exploring Late Pleistocene hominin dispersals, coexistence and extinction with agent-based multi-factor models

Quaternary Science Reviews

R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE

2022

Agents.jl: A performant and feature-full agent based modelling software of minimal code complexity

Simulation

Datseris G, R Vahdati A, DuBois T C

2022

Drivers of Late Pleistocene human survival and dispersal: an agent-based modeling and machine learning approach

Quaternary Science Reviews

R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE

2019, 221

Agents.jl: agent-based modeling framework in Julia

Journal of Open Source Software

R Vahdati, A

2019, 4(42), 1611

Population size affects adaptation in complex ways: simulations on empirical adaptive landscapes.

Evolutionary Biology

Vahdati AR, Wagner A

2017

Effect of Population Size and Mutation Rate on the Evolution of RNA Sequences on an Adaptive Landscape Determined by RNA Folding

International Journal of Biological Sciences

Vahdati A R, Sprouffske K, Wagner A

2017;13(9):1138–51

Parallel or convergent evolution in human population genomic data revealed by genotype networks

Vahdati A R, Wagner A

BMC Evolutionary Biology

2016

VCF2Networks: applying Genotype Networks to Single Nucleotide Variants data

Dall'Olio G M, Vahdati A R, Bertranpetit J, Wagner A, Laayouni H

Bioinformatics

2014

Genomic imprinting and genetic effects on muscle traits in mice

Kaerst S*, Vahdati A R*, Brockmann G, Hager R

BMC Genomics

2012

* Equal contribution

Teaching Experience

BIO397 - Applied Machine Learning - block course

University of Zurich

Lecturer

2019-2022

- I have designed and delivered lectures and exercises on Julia programming language, linear algebra, multivariate calculus, machine learning algorithms, neural networks, and evaluation and improvement of machine learning models.
- I have prepared students with no previous knowledge of machine learning to be able to design and perform complete machine learning workflows.

An introduction to agent-based modeling

University of Pusan, Busan, Switzerland

Lecturer

2022

- I gave a lecture and hands-on tutorial on using Julia language for building agent-based models.

Reproducible research - workshop

EAWAG – Kastanienbaum, Switzerland

Lecturer

2018

- I gave a lecture and hands-on tutorial on using Git version control software and collaborating using Github.

Hobbies and interests

Programming

github.com/kavir1698

2015-PRESENT

- Wrote the first agent-based modeling framework in the Julia language (`Agents.jl`) - 2019.
- Built a GUI application for organizing and quickly accessing research notes (`GraphNotes`) - 2019.
- Created a thesaurus of Persian nouns from Moeen Dictionary (`Moeen_thesaurus`) - 2018.
- Created a program to analyze photography habits using the metadata of taken photos (`AnalyzeImageMetadata.jl`) - 2018.
- Currently developing an agent-based framework for population genetics modeling.

Writing

- Writing a nonfiction book in Persian - 2021 - present.
- Created a thesaurus of Persian nouns using the Moeen dictionary - 2018.
- Translated into Persian and published Karl Popper's "Three worlds" - 2017.